

4 Lab 3: Race Cars Braking System (Brembo)

4.1 Introduction

The main objective of this laboratory is to perform a **Design Process for a Race Car Braking System**, applied to a Tatuus vehicle competing on the **Autodromo Nazionale Monza**.

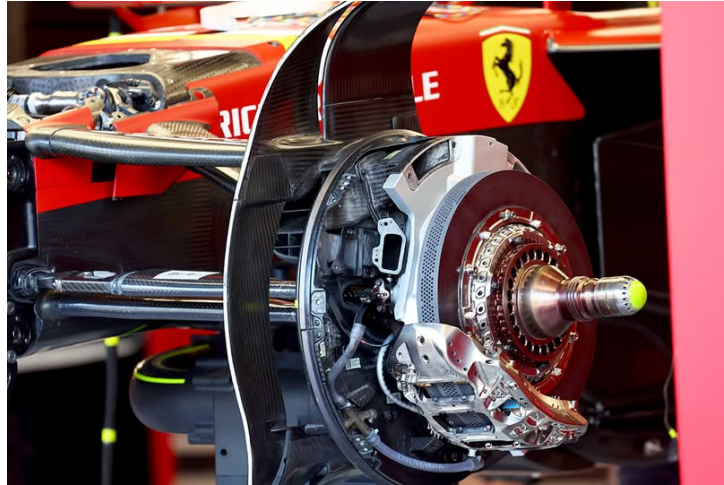


Figure 69

The design process workflow includes the following steps:

1. **Point Mass Simulation:** A preliminary single point-mass simulation on OpenLAP is launched to obtain the fundamental kinematic and dynamic data along the lap.
2. **Single-Track Model and Ideal Brake Torque Profiles:** The data extracted from the simulation serves as the input for a bicycle model application, which allows for the calculation of load transfer and the identification of the ideal braking torque limits for both the front and rear axles.
3. **Disc Thermal Sizing:** All four wheels brakes discs are sized according to their Power Dissipation Efficiency, which depends on their size and heat transfer coefficients. A constrained **Thermal Analysis** must be conducted
4. **Caliper and Master-Cylinders Sizing:** The brake circuit subsystems, such as the Brake Balance, Lever Ratios or the Calipers Sizes are designed to work in synergy with the brake discs in order to have a brake performance close to the ideal one.

4.1.1 Step 0: OpenLAP Simulation

The starting point of the analysis is a lap simulation performed using OpenLAP, a point-mass vehicle simulator. The output provides the global kinematic and dynamic behavior of the car along the track, such as the velocity profile and the global longitudinal acceleration/deceleration.. However, a point-mass model cannot provide information regarding how this braking force is distributed between the front and rear axles, which is mandatory for sizing the braking system.

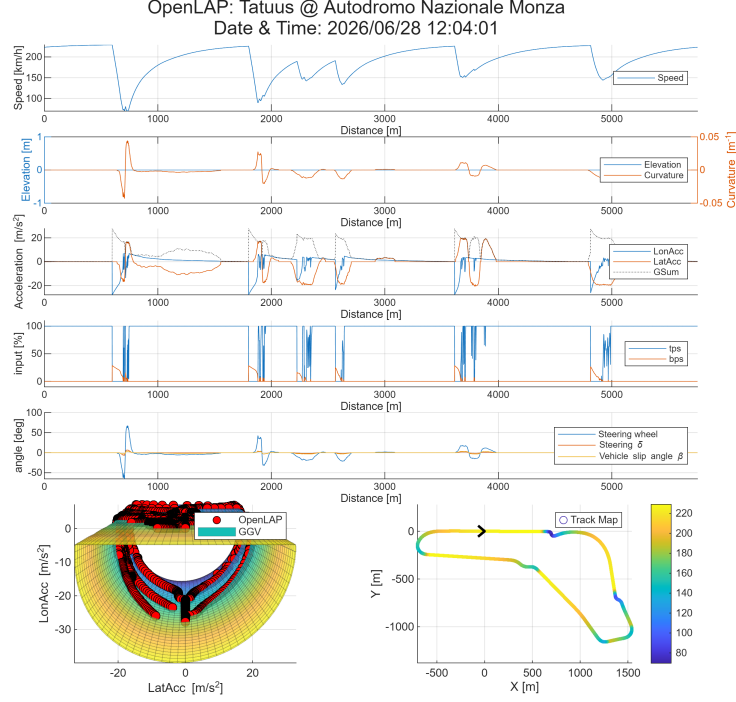


Figure 70: OpenLAP Simulation Outputs

4.1.2 Step 1: Bicycle Model and Ideal Braking Limits

To extract the ideal braking torque limits for each axle, the global data obtained from OpenLAP is processed using a single-track vehicle model.

To simplify the mathematical approach, the following assumptions are made:

- **Purely longitudinal dynamics:** The vehicle is assumed to brake only in straight lines. Steering effects, lateral forces, and trail-braking maneuvers are neglected.
- **Flat track surface:** Track slopes and banking angles are neglected.
- **Symmetric distribution:** Torque and power are split equally among the left and right wheels of the same axle.

Under these assumptions, a force balance along the longitudinal (X) and vertical (Z) axes is performed. During a braking maneuver, the vehicle experiences a longitudinal load transfer from the rear axle to the front axle due to inertia. The total vertical load on each axle is the sum of the static weight distribution, the dynamic longitudinal load transfer, and the aerodynamic downforce.

The vertical loads on the front ($F_{z,front}$) and rear ($F_{z,rear}$) axles can be expressed as:

$$F_{z,front} = m \cdot g \cdot \frac{l_r}{L} + m \cdot |a_x| \cdot \frac{h_{CoG}}{L} + F_{aero,front}$$

$$F_{z,rear} = m \cdot g \cdot \frac{l_f}{L} - m \cdot |a_x| \cdot \frac{h_{CoG}}{L} + F_{aero,rear}$$

Assuming the longitudinal tire grip is always fully saturated (i.e., operating at the limit of adherence), the maximum ideal braking force available on each axle is directly propor-

tional to its vertical load and the tire-road friction coefficient (μ):

$$F_{x,max}^{axle} = F_{z,axle} \cdot \mu$$

Finally, knowing the effective rolling radius of the wheels (R_{wheel}), the ideal maximum braking torque that each axle can sustain without locking the tires is computed as:

$$T_{brk,max}^{axle} = F_{x,max}^{axle} \cdot R_{wheel}$$

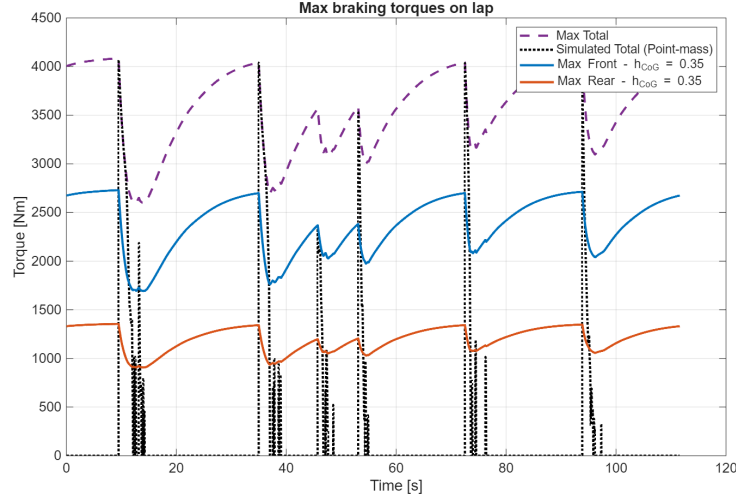


Figure 71

These ideal torque curves define the absolute physical limits of the vehicle under deceleration and represent the benchmark against which our braking system will be sized in the following exercises.

4.2 Exercise 1: Disc Thermal Sizing

4.2.1 Objectives and Requirements

The selection of the front and rear axle brake discs is driven by a strict thermal constraint: to guarantee optimal friction and braking performance, the temperature of the carbon/metallic discs must be maintained within a specific operating window, strictly between $[200^{\circ}C - 600^{\circ}C]$. Furthermore, the thermal sizing must be robust enough to withstand a multi-lap scenario, ensuring the constraints are met for a stint of at least 10 consecutive laps on the Monza circuit.

4.2.2 Braking Power Calculation

First, it is necessary to evaluate the actual energy input to the discs. From the previously calculated ideal torque limits, the sections relating exclusively to braking maneuvers are isolated, discarding the rest of the track data where the brakes are not engaged.

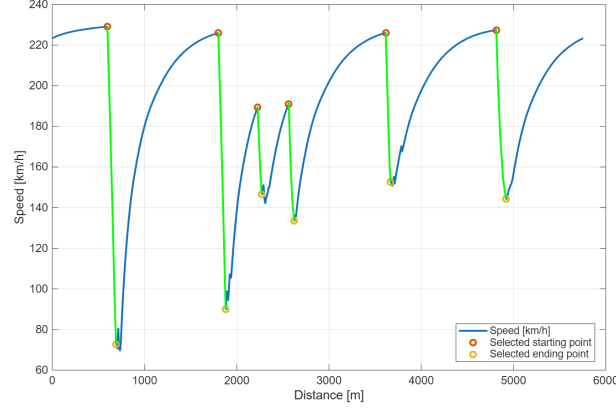


Figure 72: Braking Profiles in Monza

The maximum admissible braking power acting on a single disc (P_{brk}^{disc}) is computed as the product of the wheel's rotational speed (ω) and half of the total axle torque (T_{brk}^{axle}):

$$P_{brk}^{disc} = \omega \cdot \frac{T_{brk}^{axle}}{2} \quad (35)$$

This calculation represents the "worst-case scenario," assuming the vehicle is braking at its maximum longitudinal physical limit during every braking zone.

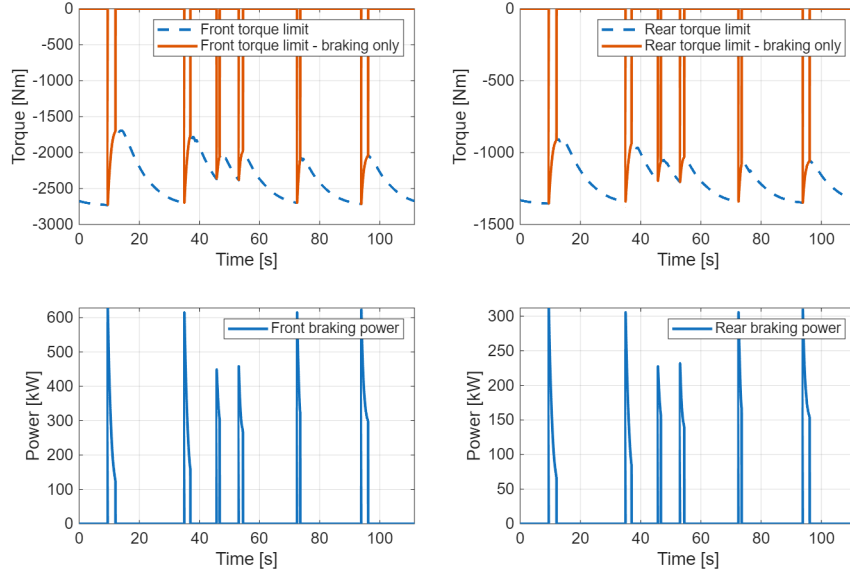


Figure 73: Braking Power Profiles

4.2.3 Thermal Model Implementation

The core of the disc sizing process relies on a thermal model. The disc temperature evolution depends on the balance between the heat generated by the braking power and the heat dissipated into the environment through forced convection and thermal radiation.

The thermal balance is given by:

$$\dot{W}_{net} = \dot{W}_{brk} - \dot{W}_{conv} - \dot{W}_{rad} \quad (36)$$

Where:

- $\dot{W}_{brk}$ is the input braking power calculated previously.
- $\dot{W}_{conv} = htc(v_x) \cdot A_{conv} \cdot (T - T_\infty)$ is the heat dissipated by convection, with $htc(v_x)$ being the speed-dependent heat transfer coefficient.
- $\dot{W}_{rad} = \sigma \cdot \epsilon \cdot A_{rad} \cdot (T^4 - T_\infty^4)$ is the heat dissipated by radiation, modeled via the Stefan-Boltzmann law.

The temperature at the next time step (T_{k+1}) is calculated from the current temperature (T_k) and the net power, considering the disc's mass (m) and specific heat capacity (c_p):

$$T_{k+1} = T_k + \left(\frac{\dot{W}_{net,k}}{m \cdot c_p} \right) \cdot \Delta t \quad (37)$$

The computational logic is implemented in MATLAB as follows:

```

1 %% Euler's method
2 for i=1:length(time)-1
3     htc_interp(i) = interp1(Vbpt,HTC,V(i),'pchip','extrap');
4
5     %Thermal Model
6     Wconv(i) = htc_interp(i) * Aconv * (T(i) - Tinf);
7     Wrad(i) = sg * epsilon * Arad * (T(i)^4 - Tinf^4);
8     Wnet(i) = Wbrk(i) - Wconv(i) - Wrad(i);
9
10    %Update
11    dT_dt(i) = Wnet(i)/(M*cp); %[K/s]
12
13    T(i+1) = T(i) + dt * dT_dt(i);
14
15 end
16
17 out.T = T - ZeroDegC_in_K; % [degC] Output temperature in degC
18 out.Wbrk = Wbrk; % [W] Braking power
19 out.Wconv = Wconv; % [W] Convection power
20 out.Wrad = Wrad; % [W] Radiant power
21 out.Wnet = Wnet; % [W] Net power

```

4.2.4 Disc Temperature Profile Extraction

The set of components provided for the sizing process includes three brake disc options with increasing geometric and thermal properties:

Parameter	Disc 1	Disc 2	Disc 3
Diameter [mm]	250	270	290
Specific heat [J/(kg K)]	1000	1000	1000
Mass [kg]	1.5	2.1	3.0
Convection area [m ²]	0.3	0.42	0.6
Radiant area [m ²]	0.07	0.07	0.07
Heat transfer coeff. [W/(m ² K)]*	75 – 125	105 – 175	150 – 250
Pad height [mm]	40	45	45

Table 15

* Evaluated for vehicle speeds $v_x \in [120 \rightarrow 300]$ km/h.

The Temperature profiles of the front and brake discs are plotted along the whole race.

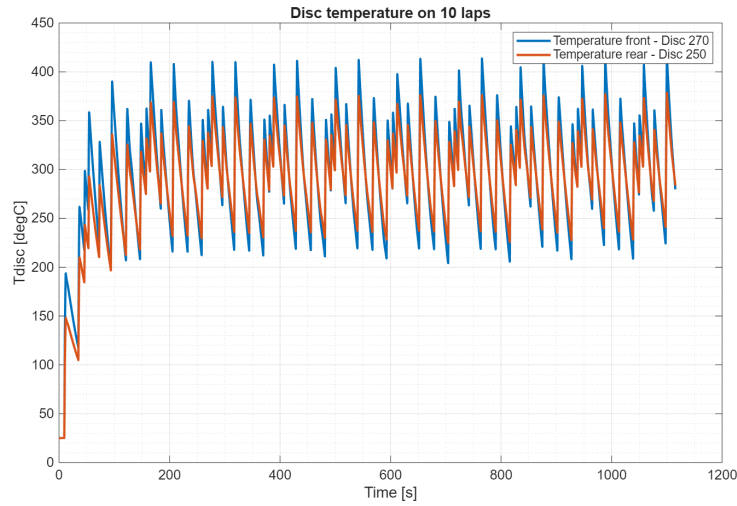


Figure 74: Braking Profiles in Monza

The final optimal set configuration mount the following discs:

Disc	Diameter [mm]	T max [°C]	T min [°C]
Front	270	416	204
Rear	250	378	225

Table 16

From the perspective of thermal analysis, varying the diameter introduces two effects on the temperature profiles.

- During heavy braking zones, a larger mass increases the overall thermal inertia of the body ($m \cdot c_p$), drastically reducing the positive thermal gradient (dT/dt) and lowering the maximum temperature peaks.
- Simultaneously, during the release phases along the straights, the larger convective area (A_{conv}) and the increased htc accelerate the rate of heat dissipation

4.3 Exercise 2: Brake Circuit Sizing

4.3.1 Objectives and Requirements

The brake mechanical input must be appropriately split into two distinct hydraulic circuits (front and rear) to achieve the so-called **Brake Balance (BB)**.

The design phase of the front and rear braking circuits involves the search of the optimal:

- Caliper Pistons: number and diameters
- Master Cylinder diameters

The goal is to implement a real brake balance that closely matches the ideal one across the vehicle's entire speed range.

This sizing process must strictly satisfy the following engineering and safety constraints:

- **Vehicle Stability (No Rear Lock):** Locking the rear wheels before the front wheels leads to a loss of directional stability (oversteer). Therefore, the rear braking force must never exceed its maximum saturation limit.
- **Hydraulic Limits:** The maximum fluid pressure in both the front and rear circuits must not exceed 120 bar.
- **Driver Ergonomics:** Since the race car is unassisted (no brake booster), the driver must be able to reach maximum deceleration applying a pedal force F_{driver} of no more than 100 kg (981 N). Ideally, this effort should remain below 80 kg to prevent driver fatigue over the course of the race.

4.3.2 Ideal Brake Balance Computation

The reference target for our design is the ideal Brake Balance, defined as the ratio of the maximum front longitudinal force to the total maximum longitudinal force:

$$BB_{ideal}(v) = \frac{F_{x,front,max}}{F_{x,front,max} + F_{x,rear,max}}$$

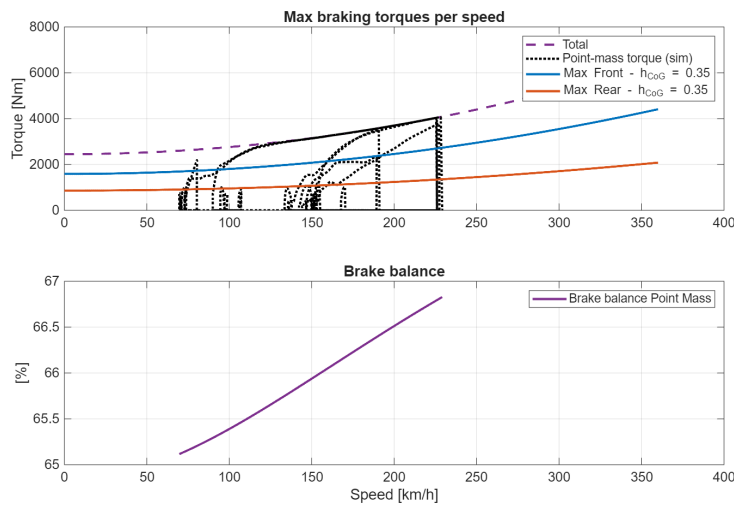


Figure 75: Ideal BB Profile

Since aerodynamic downforce and load transfer vary quadratically and linearly with velocity, respectively, the ideal brake balance is not a constant scalar but rather a function of the vehicle speed.

4.3.3 Brake System Mechanical Model

To calculate the actual braking force generated by a specific component configuration, we model the entire mechanical and hydraulic chain from the pedal to the tire contact patch.

First, the pedal box acts as a lever with a mechanical advantage K_{pedal} , defined by its geometry (l_1 and l_2):

$$K_{pedal} = \frac{l_1 + l_2}{l_2}$$

The driver's amplified force is then split between the front and rear master cylinders through a balance bar, whose distribution ratio K_{bb} depends on the pivot position (a and b):

$$K_{bb} = \frac{b}{a + b}$$

The hydraulic pressure generated in the front circuit (p_{front}) is therefore:

$$p_{front} = \frac{K_{pedal}}{A_{MCyl,front}} \cdot K_{bb} \cdot F_{driver}$$

where $A_{MCyl,front}$ is the cross-sectional area of the front master cylinder. An analogous equation applies to the rear circuit using $(1 - K_{bb})$.

At the wheel end, this pressure acts on the caliper pistons. For a multi-piston caliper, the total piston area is $A_{pist} = \sum_i \pi (D_{pist,i}/2)^2$. The clamping force generates a friction force on the disc via the brake pads, which creates a braking torque T_{brk}^{wheel} acting at the effective disc radius ($R_{discEff}$):

$$T_{brk}^{wheel} = (p_{front} \cdot A_{pist} \cdot \mu_{pad}) \cdot R_{discEff}$$

Finally, dividing the torque by the rolling radius of the tire (R_{wheel}) gives the longitudinal force at the single wheel. Since each axle has two wheels, the total actual longitudinal braking force for the front axle is:

$$F_{x,front} = 2 \cdot \frac{A_{pist} \cdot \mu_{pad} \cdot R_{discEff}}{R_{wheel}} \cdot \frac{K_{pedal}}{A_{MCyl,front}} \cdot K_{bb} \cdot F_{driver}$$

4.3.4 Component Selection

The mechanical model of the hydraulic circuit is governed by the design parameters selected within the configuration script. By iteratively tuning these parameters, we can shape the real brake balance of the vehicle. The selection is subject to specific structural and assembly constraints:

- **Master Cylinders:** Available diameters are 17, 18, 19, and 20 mm.

- **Calipers:** The front axle can accommodate 4 or 6-piston calipers, while the rear axle can accommodate 2 or 4-piston calipers. As a structural rule, the front caliper cannot have fewer pistons than the rear.
- **Balance Bar Repartition (K_{bb}):** The mechanical pivot divides the driver's effort. While adjustable between 0 and 100, optimal race car design dictates keeping this parameter within a balanced 40%–60% range.

4.3.5 Brake Balance Tuning Strategy

While the Ideal BB curve is a monotonically increasing function of speed, the Real BB is represented by a constant horizontal line. This is because the real Brake Balance is a fixed mechanical ratio defined strictly by static geometric parameters (pedal leverage, master cylinder areas, piston areas, and balance bar setting). It has no active mechanism to adapt to the vehicle's speed.

Therefore, the challenge consists in finding a constant Real BB line that sits in the perfect "sweet spot" relative to the Ideal BB curve. The tuning strategy follows these rules:

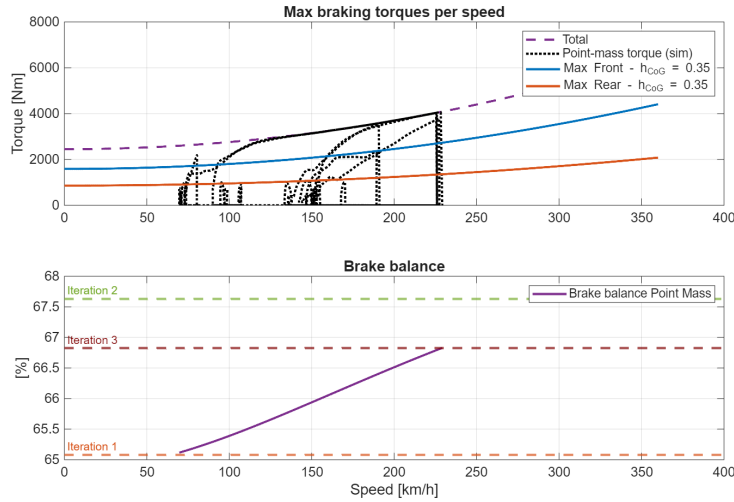


Figure 76: Real BB Applications

- **Below the Ideal Curve (Iteration 1):** If the Real BB falls below the ideal curve, the system sends a higher proportion of braking force to the rear tires than they can physically sustain. This inevitably leads to premature rear-wheel lock-up, causing severe oversteer and total loss of vehicle stability.

Parameter	Configuration
MC Diameter (Front / Rear)	17 mm / 17 mm
Front Caliper (Pistons × Diameters)	4 × [30, 32] mm
Rear Caliper (Pistons × Diameters)	4 × [26] mm
Balance Bar Repartition	45%

Validation Parameter	Result
Max Front Pressure	82 bar
Max Rear Pressure	67 bar
Max Driver Force	849 N (87 kg)
Front Brake Balance	65.08%
Rear Brake Balance	34.92%

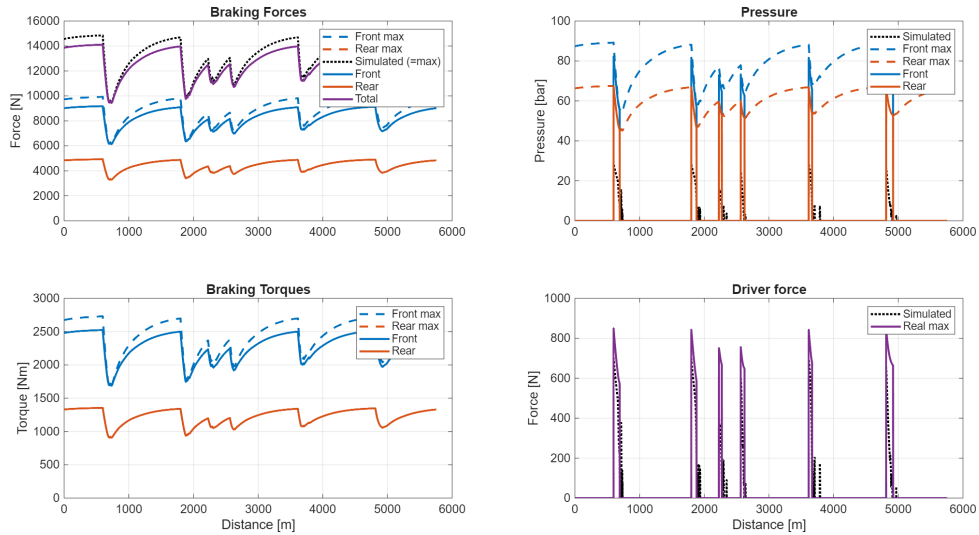


Figure 77: Iteration 1

- **Above the Ideal Curve (Iteration 2):** If the Real BB is placed too high above the ideal curve, the system heavily biases the front tires. While this configuration is dynamically stable, it heavily under-utilizes the rear tires' grip capacity.

Parameter	Configuration
MC Diameter (Front / Rear)	17 mm / 18 mm
Front Caliper (Pistons × Diameters)	4 × [30, 32] mm
Rear Caliper (Pistons × Diameters)	4 × [26] mm
Balance Bar Repartition	45%

Validation Parameter	Result
Max Front Pressure	89 bar
Max Rear Pressure	65 bar
Max Driver Force	918 N (94 kg)
Front Brake Balance	67.63%
Rear Brake Balance	32.37%

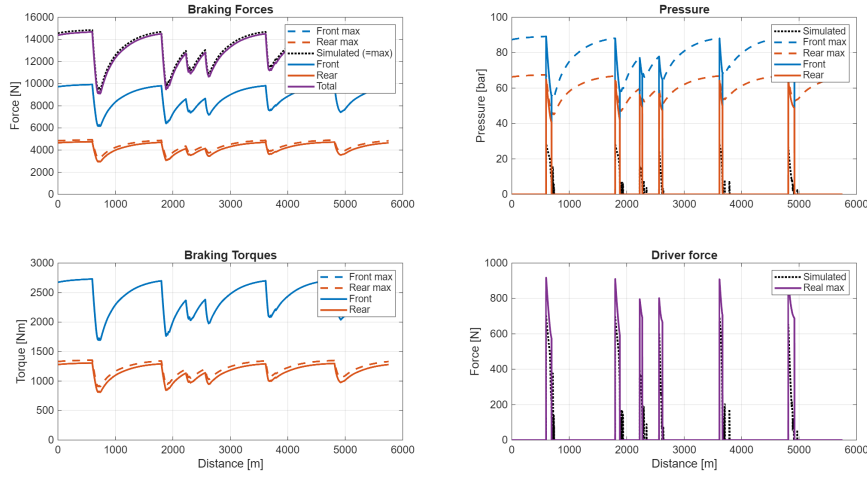


Figure 78: Iteration 2

- **The Optimal Target (Iteration 3):** the constant BB guarantees that the front wheels will always lock slightly before the rear wheels (ensuring safety and stability), while maximizing the grip exploitation on both axles for the shortest possible stopping distance.

Parameter	Configuration
MC Diameter (Front / Rear)	17 mm / 17 mm
Front Caliper (Pistons × Diameters)	$6 \times [30, 32, 32]$ mm
Rear Caliper (Pistons × Diameters)	$4 \times [28, 28]$ mm
Balance Bar Repartition	50%

Validation Parameter	Result
Max Front Pressure	58 bar
Max Rear Pressure	58 bar
Max Driver Force	659 N (67 kg)
Front Brake Balance	66.83%
Rear Brake Balance	33.17%

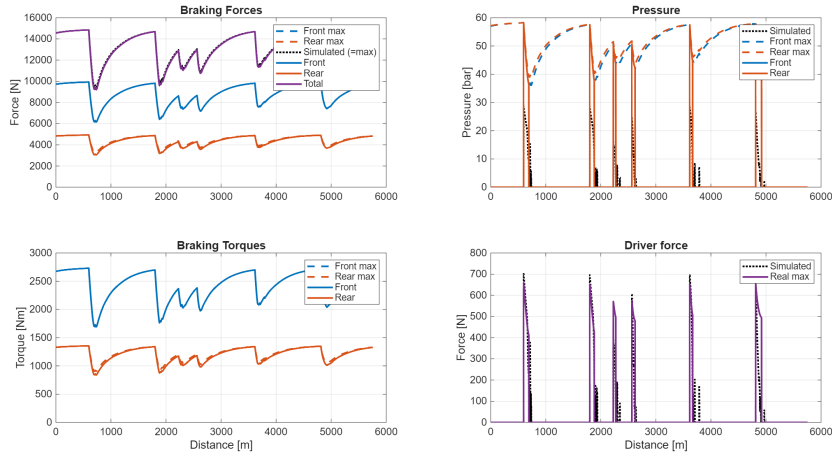


Figure 79: Iteration 3

4.4 Exercise 3: Bicycle Model simulation

The aim of this exercise is to evaluate how the real brake system affects the lap time performance. While the point-mass model can exploit an ideal brake force distribution at each instant, the bicycle model uses a real hydraulic brake system with fixed components and fixed brake balance.

Configuration	Front Caliper	Rear Caliper	MC F/R	Balance Bar
Baseline	$6 \times [30,32,32]$	$4 \times [28,28]$	17/17 mm	50%
Alt. 1 - Front biased	$6 \times [30,32,32]$	$4 \times [28,28]$	17/17 mm	40%
Alt. 2 - Rear biased	$6 \times [30,32,32]$	$4 \times [28,28]$	17/17 mm	60%
Alt. 3 - Conservative	$6 \times [30,32,32]$	$4 \times [26,26]$	17/17 mm	50%

Table 17 Brake system configurations tested in the bicycle model.

Alternative 1 was selected to represent a front-biased setup. This configuration is expected to be stable but suboptimal, since the front axle reaches the grip limit before the rear axle is fully exploited.

Alternative 2 was selected to represent a rear-biased setup. This configuration is expected to be critical from a stability point of view, since the rear axle may saturate before the front axle.

Alternative 3 was selected as a conservative setup. The rear caliper piston area is reduced, increasing the safety margin against rear locking but reducing the exploitation of the rear axle.

Simulation	Lap Time [s]	Δt vs PM [s]	Brake Balance	Max Driver Force [kg]
PointMass	111.54	0.000	-	57.594
SingleTrack Baseline	112.75	1.2048	0.62579	57.492
Alternative 1	112.94	1.3913	0.73823	51.478
Alternative 2	112.92	1.3706	0.55623	48.012
Alternative 3	112.82	1.2774	61.672	0.68558

Table 18 Lap time comparison between point-mass and bicycle model simulations.

The point-mass simulation represents the ideal reference, since it can always exploit the optimal brake force distribution. All bicycle model simulations show a lap time penalty because the brake force distribution is constrained by the real hydraulic system.

The realistic brake system affects lap time because it cannot continuously adapt the front/rear brake force distribution as the ideal point-mass model does. A fixed brake balance inevitably introduces a performance penalty. Among the tested real configurations, the baseline setup obtained from Exercise 2 provides the best compromise between performance and stability, since it remains close to the ideal brake balance while avoiding rear locking.

